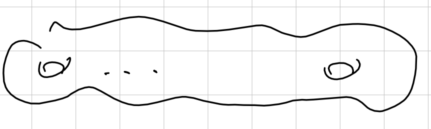
$S^n$, ~~$S^1\times S^1$~~, $\mathbb{RP}^n$,



($n\geq1$

↓

$\pi_1=0$.

2-dim mfd

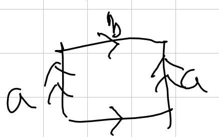
$L(p,q)$

$M\#N$. ∧ ∨

$S^1\times S^1/e^{2\pi i p}, e^{2\pi i q}$

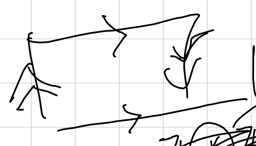
$[0,1]^2$

Klein.



torus.

$\mathbb{Z}\times\mathbb{Z}$



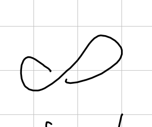
Klein. free product.

~~$\mathbb{Z}\times\mathbb{Z}_2$~~

$ababab b$.

$\langle a,b \mid aba=b\rangle$.

$aba^{-1}b^{-1}=e$.

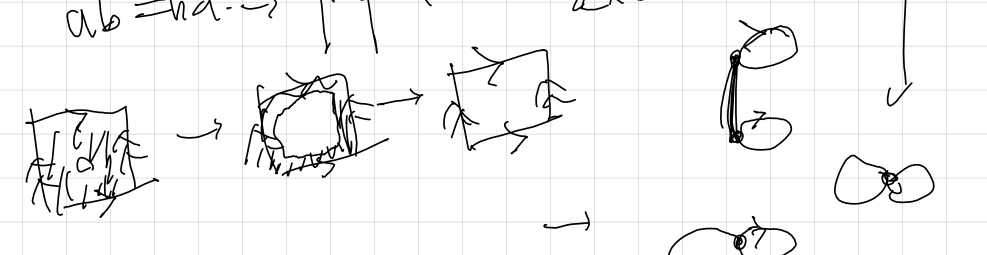


$\langle a,b \mid ab=ba\rangle$

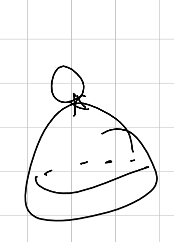
$\mathbb{Z}\times\mathbb{Z}$

$S^1\vee S^1$

$ab=ba \to$ group +Z.



$S^1\vee S^2$



$T: \mathbb{Z}\times\mathbb{Z}$

$S^1\vee S^1$

$\mathbb{Z}*\mathbb{Z}$

$T\setminus\bullet$ $\mathbb{Z}*\mathbb{Z}$

$K\setminus\bullet$ $\mathbb{Z}*\mathbb{Z}$

theorem

Theorem

$S^1\vee S^1$

$\mathbb{Z}*\mathbb{Z}$

$S^1$ ↕ $\mathbb{Z}$

$S^3: 0$.

↓

$\mathbb{R}^3: 0$.

$M$: n-dim with $n\geq3$

then $\pi_1(M)\simeq\pi_1(M\setminus\bullet)$.

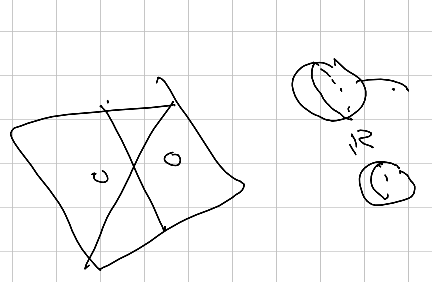
$\simeq\pi_1(M\setminus 2pt)$.

$\simeq\pi_1(M\setminus\{n\})$.

$\pi_1(S^n)=0$ $n\geq2$.

$\pi_1(\mathbb{RP}^n)=\mathbb{Z}_2$ ($n\geq2$).

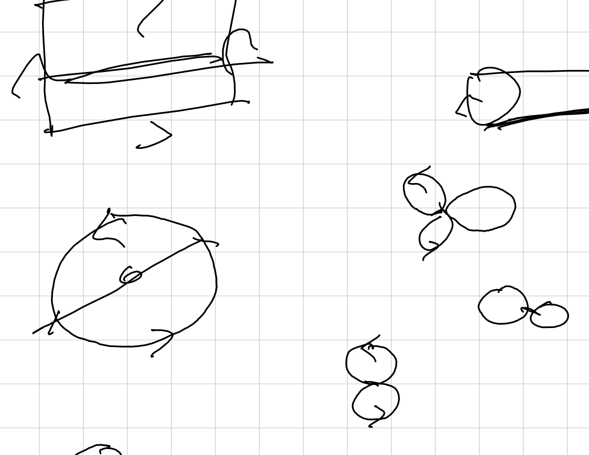
tree



~~$S^1\vee S^1\vee\mathbb{R}$.~~

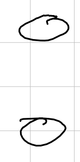
↓

$\mathbb{Z}*\mathbb{Z}$

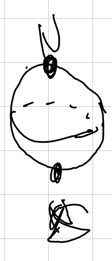


: $S^1\vee S^1\vee S^1$

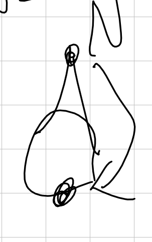
$\mathbb{Z}*\mathbb{Z}*\mathbb{Z}$

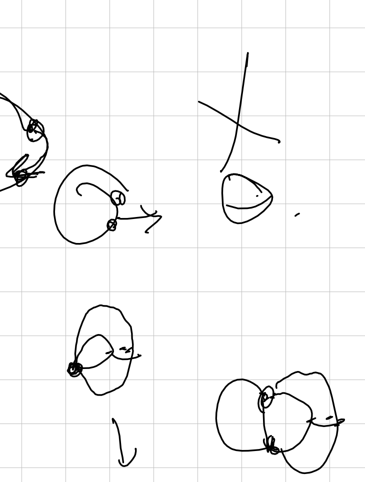


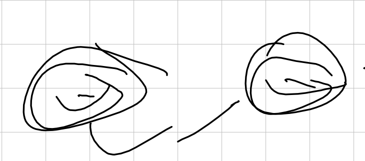
Example.



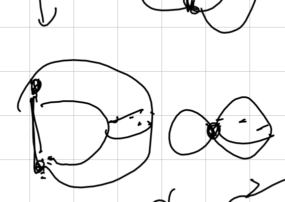
$S^2/N=S$



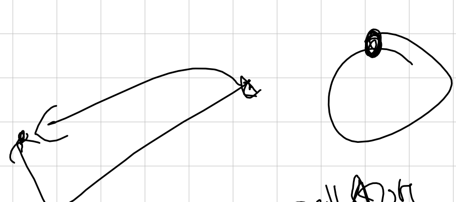




$\to\pi_1(\ldots)$



$S^1\vee S^1$



$S^1$: 0-cell 1개

1-cell 1개.

$S^2$: 0-cell 1개

2-cell 1개

$S^1\vee S^1$

$\mathbb{RP}^2$: 0-cell: 1개

1-cell: 1개

2-cell 1개

$aba^{-1}b^{-1}=0$.

$\langle a,b \mid ab=b\ldots\rangle$

$(\mathbb{Z}*\mathbb{Z})/\alpha$

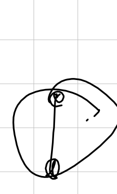
$\mathbb{RP}^2: \mathbb{Z}_2$ $\langle a \mid a^2\rangle$

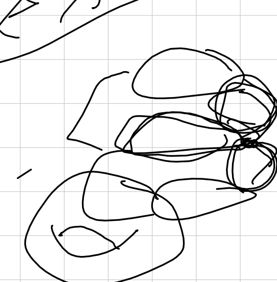
$a^2=1$

$\mathbb{RP}^3$: ~~0-cell 1개 1-cell 1개 2-cell 1개 3-cell 1개~~

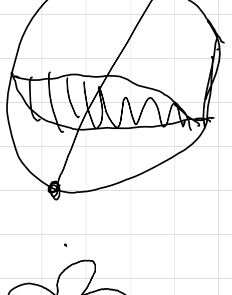
~~$\pi_1(\ldots)$~~

$\pi_1(\mathbb{RP}^3)=\pi_1(\mathbb{RP}^2)=\mathbb{Z}_2$.





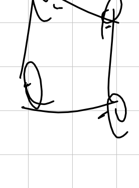
$(\frac12,\frac12,\frac12)$

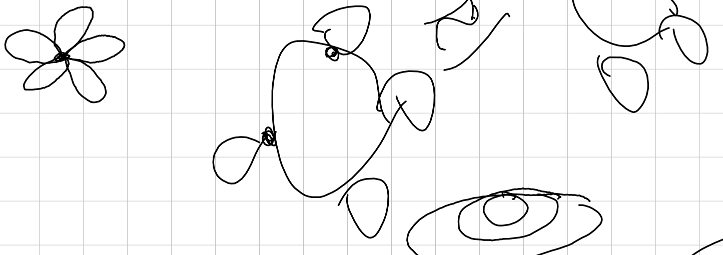


← $x^2+y^2+z^2=1$

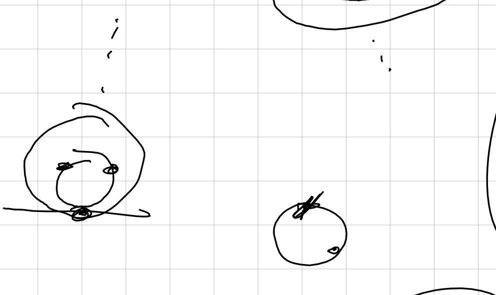
∪ $x^2+y^2\leq1$

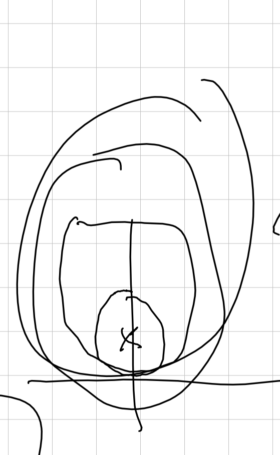
∪ $(0,0,z)$ $-1\leq z\leq1$.

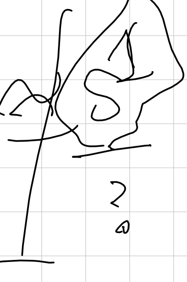


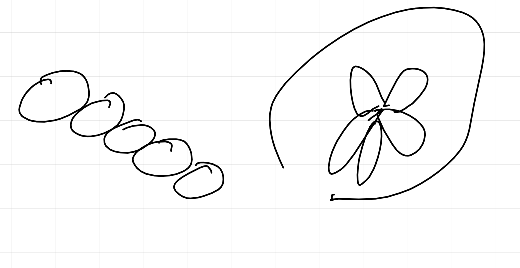


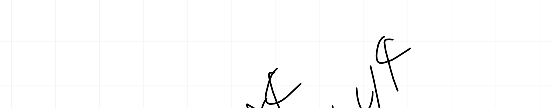
$\mathbb{R}/\mathbb{Z}$











$z=x^4+y^4$

↓ exp

$P=0$

$K_1=f(y,z)$

$K_2=f(y\ldots$

$P_n+z$.

$\int\frac{\ldots}{y}$ $\int\frac{1}{z}dz$

$4$

$2\pi i$.

$2\pi i\cdot X(\ldots)$

$0$